MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : PEC-IT601D Image Processing

UPID : 006591

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(I) What is the primary purpose of contrast stretching?

(II) The _____ approach in image restoration involves using mathematical equations to estimate the original image.

(III) The _____ operator is a gradient-based technique used to detect edges by computing the derivative of an image.

(IV) Quantization is digitizing the _____ value.

(V) The process of converting an analog image into a digital image is called _____.

(VI) A pixel's immediate neighbors in a 4-connectivity system are positioned in the ____, ____, ____, and ____ directions.

(VII) Contrast enhancement techniques modify an image's _____ to improve visibility.

(VIII) The degradation model in image restoration is mathematically represented as: _____

(IX) _____ is a local processing technique used to connect edge pixels.

(X) An 8-bit grayscale image can have a maximum of_____ shades of gray.

(XI) What happens when the scaling factor is less than 1?

(XII) Given two pixels at coordinates (3,5) and (7,9), what is their chessboard distance?

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. a) State the condition of transformation function s=T(r). [5]
b What is gamma correction?
c) A captured image appears very dark because of wrong lens aperture setting. Which enhancement technique is appropriate to enhance such an image?

3. List down various Geometric Transformation. Differentiate between unconstrained and constrained restoration approaches. [5]

4. Write a short note on Hough Transformation. [5]

5. What are the different types of image storage formats? [5]

6. Describe the point spread function (PSF) and its role in image degradation. [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. Describe in detail the elements of digital image processing with Diagram. [15]

8. (a) Describe a simple Image Model. [5]
(b) What is perspective projection? Explain in details. [5]
(c) Compare rod cell and cone cell in details. [5]

9. (a) Define arithme tic opera tions (addi tion, subtrac tion, mul tiplication, division) and their effects on images. [5]
(b) Explain logical operations AND, OR, XOR and NOT with example. [5]
(c) Derive the Euclidean, City-block, and Chessboard distance formulas and compare their applications in image analysis. [5]

10. (a) Given an image with the following histogram: [7]

Intensity Level012 345

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Pixel Count 5912768

And a desired histogram:

Intensity Level01 2 345

Pixel Count 41010869

Perform histogram matching and find the new intensity mapping.

(b) Suppose a digital image is subjected to histogram equaliza tion. What effect will a second pass equalization have over the equalized image? [5]

(c) Differentiate between spatial domain enhancement and frequency domain enhancement. [3]

11. (a) What is image segmentation ? Explain global processing via Hough transform. [5]

(b) How do the relative size of object and background regions influence threshold detection? [2]

(c) What are the differences between Line and Edge Detection? [3]

(d) Why second derivative operation is not normally used or Edge detection? [2]

(e) How to choose the seed points for region growing operation? [3]

*** END OF PAPER ***

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